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The risk we bear: The effects of review speed and industry user fees on new drug safety☆

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Abstract

Drug review speed has increased substantially in the 1990s, largely due to industry-funded user fees. Following several drug withdrawals, however, new questions have emerged about the effects of this change on drug safety. This article examines the impact of review speed and user fees on counts of serious adverse reactions among drugs approved in 1990–2001. The analysis controls for the influence of drug utilization, patient conditions, drug novelty, black box warnings, foreign drug launch, US launch lags, patient age, and gender on drug reactions. Results show that drugs receiving faster reviews are associated with increased counts of serious adverse drug reactions. Other results show that novel drugs, drugs with black box warnings, drugs first launched abroad, and drugs with shorter US launch lags also have increased adverse drug reactions. Although any increase in risks must be weighed against benefits, the results show a trade-off between review speed and drug safety.
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1. Introduction

Drug safety and the speed of new drug review have generated much controversy in the debate over pharmaceutical regulation over time. In the past, long delays and lengthy review times characterized the FDA's drug approval process. This changed, however, with the 1992 Prescription Drug User Fee Act (PDUFA), which required firms to pay fees to the FDA to help finance drug reviews. Under PDUFA increasing the speed of drug review became a new priority for the agency. User fee revenues, which currently represent 40–50% of the agency's new drug review resources, have resulted in substantial reductions in drug review times.[1] At the same time, the withdrawal of several drugs in the user fee era has raised concerns about drug safety.[2] The drug withdrawals led to new questions about the effects of increased review speed and PDUFA on drug safety (Willman, 2000).

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[1] See http://www.fda.gov/oc/oms/ofm/budget/documentation.htm (3 April 2005).
[2] Twelve drugs were withdrawn from the market between 1997 and 2001: Redux, Pondimun (1997); Duract, Posicor, Seldane (1998); Hismanal, Raxar (1999); Lontronex, Propulsid, Rezulin (2000); Baycol, Raplon (2001). Lotronex was reintroduced in 2002, but under restricted distribution.

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Proponents of PDUFA argue that user fees provide extra resources to help speed review and hence accelerate patient access to new medicines. Faster access to innovative drugs may improve patient health. Critics of PDUFA argue that the reform encourages too much emphasis on speed and not enough emphasis on safety. They worry that the FDA's dependence on user fee revenue may alter its priorities in ways that could possibly harm patients.[3] This debate reflects the fundamental tension in drug regulation between the agency's competing goals of ensuring drug safety and facilitating drug access. It also highlights the need for more systematic empirical evidence about the effects of faster drug reviews and PDUFA on drug safety and patient health.

One challenge in examining these effects has been finding a good proxy for drug safety. Initially, studies focused on drug withdrawals to gain insights about drug safety. Friedman et al. (1999) found little difference in drug withdrawal rates over time, hence concluding that PDUFA did not reduce drug safety. Examining more data, GAO (2002) found that drug withdrawal rates increased after PDUFA. Tufts (2005) also found that drug withdrawal rates increased in the 1990s, but then decreased after 2000. Berndt et al. (2005) found no significant difference in drug withdrawal rates before and after PDUFA although they note that results are sensitive to the time periods selected and the potential problem of censoring. One limitation of this proxy is that drug withdrawals reflect only a small number of drugs. This makes it is difficult to draw general conclusions about drug safety. In addition, studies which focus on drug withdrawals say little about the safety of drugs that remain in the market and hence have the greatest impact on public health. The frequency of black box warnings given to drugs has also been studied by Begosh et al. (2006). They find that the rate of black box warnings given after approval was not affected by PDUFA. While black boxes occur with greater frequency than drug withdrawals, they are discretionary FDA actions (to seek a label change) that must be negotiated with firms after approval.[4] A richer measure of drug safety that applies to more drugs and does not reflect such regulatory discretion could produce more insight about the effects of PDUFA on drug safety.

Adverse drug reaction data have been studied by Olson (2002) and Olson (2004b). Olson (2002) found that faster reviews were associated with increased adverse drug reactions for the 1990–1995 drug approvals. Olson (2004b) found that drugs deemed therapeutically novel by the FDA were also associated with increased drug-related risks after approval. A study by Grabowski and Wang (2006b), which also examined adverse drug reactions, found no relationship between review speed and adverse effects among the 1992–2002 approvals, but it did find that novel drugs were associated with more adverse drug reactions than other drugs. These authors use a somewhat different approach and method for counting adverse effects than is used in Olson.[5] These different findings suggest that more research is needed about the speed–safety relationship.

There have been several changes that might affect the relationship between review speed and drug safety. Increases in the number of patients included in clinical trials combined with increases in drug review staff at the FDA in the 1990s could lead to better safety screening under PDUFA (Grabowski and Wang, 2006b), and perhaps fewer drug reactions. However, increased pressures to speed approval and meet PDUFA review deadlines could lead to reduced drug safety, and hence more drug safety problems.[6] A shift toward more first drug launches in the US reported by Grabowski and Wang (2006a) could result in fewer spillovers of safety information from abroad (from other countries experiences with a drug) prior to US approval and consequently more risks for patients who take new drugs.

This paper examines the effects of review speed, user fees, and other factors on counts of adverse drug reactions (ADRs) among new drugs. The analysis expands upon prior research by including more drug approvals (from 1990 to 2001), more adverse drug reactions (from 1991 to 2003), and additional controls for black box warnings at the time of approval, foreign drug launch prior to US approval, and launch lags to US approval. The analysis also controls for

In contrast, Bakke et al. (1995) reports that only 10 safety-related drug withdrawals were observed in the prior two decades, 1974–1993.

[3] Zelenay (2005) argues that the "FDA's user fee program inappropriately sacrifices the public's health to the pharmaceutical industry's gain". Okie (2005) suggests that the user fee program has "created a frenetic atmosphere in which the pharmaceutical industry is the customer and scientific debate is discouraged".

[4] Begosh et al. report that boxed warnings at approval apply to 16% of all approved drugs. Once drugs are on the market, FDA cannot unilaterally compel labeling changes or the addition of black box warnings.

[5] This study will examine the sensitivity of the results to the alternative counting method.

[6] In 1997 prior to the renewal of PDUFA (signed into law November 21), the FDA approved five drugs that were eventually withdrawn from the market: Duract (July), Baycol (June), Raxar (November), Posicor (June), and Rezulin (January). This corresponds to a 13.8% withdrawal rate among the 1997 drug approvals that calendar year.

drug utilization, the conditions treated, important disease characteristics, therapeutic novelty, patient age, gender, and year effects.

The primary result is that the drugs receiving faster reviews have higher counts of serious drug reactions including those resulting in hospitalization and death after approval than the drugs receiving slower reviews. This result suggests that there is a trade-off between the speed of review and reported adverse drug effects. Once the effects of review speed are taken into account, the analysis shows little association between user fees and serious ADR counts, which suggests that the primary impact of user fees or PDUFA on drug risks occurs through review speed. Results also show that novel drugs, drugs with a black box warning at the time of approval, drugs launched abroad prior to US approval, and drugs with shorter US launch lags are associated with higher counts of serious adverse drug reactions.

The next section provides some background and describes cost trade-offs in the review process. Section 3 discusses the impact of PDUFA/FDAMA on the regulatory cost trade-offs and develops hypotheses about the effects of the user fee financing and increased review speed on drug safety. Section 4 describes the data and methods for the analysis. Section 5 presents the results. Section 6 provides some concluding remarks including a discussion of limitations and policy implications.

2. Drug approval and FDA regulatory goals

The FDA is a public health, regulatory agency, which is responsible for evaluating the safety and efficacy of the nation's pharmaceuticals. Firms are required to conduct clinical trials to show that drugs are safe and effective for their intended use. After firms submit this evidence to the FDA, regulators assess whether the benefits of the drug exceed its known risks. The difficulty facing regulators is that there is always uncertainty about risks relative to benefits at the time of approval. Such uncertainty is due to the limitations of clinical trials, which include their size and duration. For example, little may be learned about rare side effects, side effects arising from long-term use, and side effects in understudied populations because clinical trials are relatively small, relatively short, and the patients studied may not be representative of the patients who end up taking the drug after approval.[7] Clinical studies also provide little information about drug interactions and the effects of drugs among the elderly or the chronically ill who are typically large users of prescription drugs, because such patients may be underrepresented in clinical studies. This uncertainty complicates the job of the regulator in making risk benefit assessments.

Uncertainty about drug effects creates the potential for two types of regulatory errors. First, regulators could approve a dangerious or ineffective drug or second, they could fail to approve a medically useful drug. There is a conflict among the agency's goals to ensure drug safety and facilitate drug access. Tough regulatory standards and careful screening can help keep some dangerous drugs off the market thus furthering the goal of safety.[8] However, tough screening also delays the introduction of innovative new medicines, which hinders the goal of access. Alternatively, agency efforts to accelerate drug access (by speeding review) may lead to the approval of some drugs that could harm consumers. In this context, the speed of approval will be determined by how regulators weigh the expected benefits from faster drug access against the expected costs from reduced safety.

Grabowski and Vernon (1983) characterized the cost trade-offs of these two errors in the pre-market testing of new drugs. This analysis extends these trade-offs to the period of pre-market review of new drugs. They are depicted in Fig. 1 as the cost of reduced drug access (benefits to patients that are foregone from lack of access) and the expected cost of exposure to dangerous drugs. The expected cost of unsafe drugs is assumed to decline as the length of review increases. With more time, reviewers can increase scrutiny and seek more information to inform drug safety, which reduces the probability of approving an unsafe drug. In contrast, the expected cost of reduced drug access increases with the length of review because longer reviews delay patients' access to beneficial new drugs and reduce a drug's effective patent life.[9] The sum of these two cost curves represents the total social cost of the length of review. Assuming that regulators want to minimize the overall social cost of their choice of r, the optimal review time is found at the minimum r_1.

[7] IOM (2006) suggests that on average, a trial involves between 600 and 3000 patients. Some trials may be either smaller or larger.

[8] In a comparison of drug withdrawals in the US and UK in 1971–1992, Abraham and Davis (2005) show that FDA's lengthy review times and greater regulatory scrutiny prevented the approval of some unsafe drugs in the US compared to the U.K. and that US regulators identified in pre-market testing the same safety problems upon which drug withdrawal ultimately occurred in the UK.

[9] Vernon et al. (2006) show that faster reviews increase firm incentives for R&D.

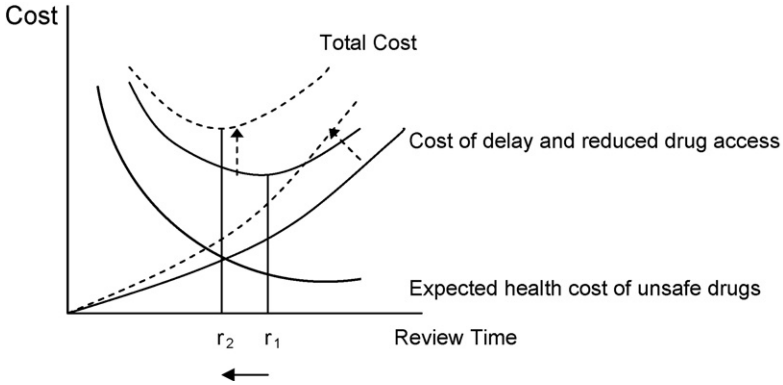

Fig. 1. Costs trade-offs in new drug review.

Deviations from the optimal review time may occur if the agency either overestimates the expected costs of unsafe drugs or underestimates the costs of reduced drug access. Quirk (1980) argues that drug tragedies create large political costs for the agency, which could raise the agency's estimate of the costs of unsafe drugs. With the exception of AIDS drugs, there have been few political consequences for the FDA for failing to approve beneficial drugs. That changed in the 1990s as AIDS activists successfully pressured the FDA to accelerate the approval of new AIDS drugs (Epstein, 1996). The pressure from AIDS activists created new political costs for delaying the approval of AIDS drugs and consequently led to new efforts to accelerate their approval. The success in accelerating the approval of AIDs drugs subsequently led to new efforts and legislation to accelerate the approval of all drugs.

3. Legislation to reduce review times: PDUFA and FDAMA

The Prescription Drug User Fee Act (PDUFA) was signed into law on October 29 1992 to speed up the review process. PDUFA gave the agency additional resources to help fund review activities, and in return the agency was expected to accelerate the review process and meet new review deadlines: 6 months for priority-rated drug applications (drugs deemed therapeutically novel) and 12 months for other drug applications. PDUFA also contained annual revenue targets for the FDA and a schedule of user fees designed to meet those targets.[10] Fee revenues were designed to supplement existing congressional appropriations.[11] To help ensure agency accountability, FDA was required to report annually to Congress on the status of meeting the review deadlines and other performance goals as well as detail how user fee revenues were spent.

The Center for Drug Evaluation and Research (CDER) received most of the user fee revenues. A majority of fee revenue was used to hire new reviewers and program managers for drug review teams, while a smaller portion was used to introduce administrative reforms to improve application tracking.[12] Under PDUFA, fee revenues could only be used to accelerate drug review and could not be used for post-market drug safety monitoring (until 2002) or for other activities such as generic-drug review, medical-device review, or food regulation.[13] PDUFA also limited the extent to which Congress could reduce agency appropriations in response to increasing user fee revenue. Finally, PDUFA had a fixed 5-year term so that its stakeholders could revisit the program prior to its renewal. Without new legislation, PDUFA would sunset at the end of 5 years.

The fixed term meant that the FDA needed to make a case for renewal at the end of 5 years. Since PDUFA reauthorization would bring industry, back to the bargaining table, FDA needed to show that they were meeting the

[10] Fees for new drug applications were listed as $100,000 in 1993, $150,000 in 1994, $208,000 in 1995, and $233,000 in 1996. Fees could be adjusted in any year to meet the inflation-adjusted revenue targets.

[11] The agency collected fee revenues of $36 million in 1993, $56 million in 1994, $78 million in 1995, $85 million in 1996, $88 million in 1997, $118 million in 1998, $126 million in 1999, $141 million in 2000, and $139 million in 2001 (FDA, 2005).

[12] There were 561 PDUFA-funded FTEs in CDER in 2000. Overall CDER staff increased from 1408 FTEs in 1992 to 1777 FTEs in 2000. There were 201 PDUFA-funded FTEs in CBER in 2000. Overall CBER staff increased from 643 FTEs to 802 FTEs in 2000.

[13] In 2002, PDUFA III allowed for a small amount of fee revenues to be used for safety surveillance of a drug in its first 2 or 3 years on the market. Fee revenues could not be used for other post-market safety actions or safety studies.

review deadlines to justify continued user fee funding. This strengthened the incentives in CDER to reduce delay and meet drug review deadlines (Olson, 2004a). Olson suggests that these incentives along with the new resources were instrumental in reducing delays.[14]

In terms of Fig. 1, the threat of non-renewal (i.e. the loss of industry user fees) created new political consequences for delaying drug approval. These political consequences altered the cost trade-offs facing regulators since both political and social costs impact agency decisions.[15] The new political consequences of PDUFA effectively raised the agency's estimate of the cost of delay, which in turn results in an upward rotation in the cost of delay curve facing the agency. As the social and political cost of delay increase, the regulator's cost-minimizing review time falls to r_2. If increased user fee funding were directed toward better detection of post-marketing drug safety problems, it could also lead to a downward shift in the expected health costs from dangerous drugs. However, since the reform prevented any user fee funds from being used for post-marketing surveillance (until 2002), no such effect is expected during this period. PDUFA funds could have also been used to improve safety screening in the pre-market. Since the performance goals only relate to speed not safety, there is little evidence to gauge such an effect. Whether or not such an effect exists is an empirical question.[16]

The figure shows two important effects of the reform. The first is a reduction in review times, which has been well documented (FDA, 1997; CDER, 2000; Olson, 2002, 2004a). The second effect is an increase in the amount of expected costs from unsafe drugs at the new optimum. These direct effects are the basis for Hypotheses 1 and 2.

Hypothesis 1. Reductions in drug review times lead to increases in drug-related risks for patients.

The pressures to approve new drugs might also lead regulators to approve some drugs that might not otherwise have been approved independent of the speed of review. Alternatively, firm behavior regarding drug submissions may have changed in the user fee era. Anticipating a greater chance of approval, firms may choose to submit riskier drugs than in the past. The second hypothesis examines the effect of PDUFA on drug-related risks controlling for the speed of new drug review. By examining the effects of PDUFA and the speed of new drug review simultaneously, the analysis can determine whether increases in drug-related risks in the PDUFA era are due solely to increases in drug review speed or to other reform-related factors.

Hypothesis 2. User fee funding has led to an increase in drug-related risks even after controlling for the effects of review speed.

If political pressures to speed drug review were coupled with reduced regulatory stringency, then reductions in the drug review times could have a greater effect on drug-related risks in the user fee era. This would correspond to a positive interaction term for review speed and PDUFA. Alternatively, if user fee funds were associated with improved safety screening, then we might expect a smaller effect of review time on drug risks in the user fee era. This would correspond to a negative interaction term for review speed and PDUFA. Pressures to accelerate approval were greatest for therapeutically novel drugs. If these pressures led regulators to approve a more risky set of novel drugs, then the effect of drug novelty on drug risk may increase in the user fee era. Alternatively, if PDUFA was associated with improved safety screening among novel drugs, then the effect of drug novelty on drug risk may diminish in the user fee era. Hypotheses 3 and 4 are potential interaction effects among PDUFA and review speed and among PDUFA and drug novelty.

Hypothesis 3. The effect of new drug review speed on drug-related risks is significantly different in the user fee era.

Hypothesis 4. The risks associated with therapeutically novel drugs are significantly different in the user fee era.

[14] Olson's analysis showed that the observed decline in review times could not be fully explained by the increase in CDER resources (review staff) alone. Approximately 34% of the observed decline in review times was explained by other reform-related factors besides increased resources.

[15] Thompson (2000); see quote from J. Woodcock.

[16] By examining how the speed of review relates to ADR counts after approval, we may be able to gain some insight into the answer to this question.

4. Data and methodology

The empirical analysis examines the impact of drug review times, PDUFA, and other factors on adverse drug reactions for new chemical entities approved in the calendar years of 1990–2001. This period is particularly interesting to study because it corresponds to the years in which the pressures to accelerate review times markedly increased and drug safety issues among new drug approvals were beginning to emerge. This period also includes drugs approved both before and after PDUFA so that review times will have considerable variation, which is needed to examine the speed–safety trade-off.

4.1. Adverse drug reactions

Adverse drug reaction (ADR) data is used to measure drug-related risks. These data come from the FDA's Adverse Event Reporting System (AERS), which is the agency's database for monitoring and tracking new information about drug risks after FDA approval. The ADR data represent an improvement over drug withdrawal data because they provide risk information about all newly approved drugs. Also, these data are an improvement over drug poisoning data (Peltzman, 1987) because they offer a broader range of serious drug reactions, not just ones which result in death.

ADR reports may be submitted by physicians, pharmacists, clinicians, and patients. However, most ADR reports originate from health professionals whose patients experience serious reactions to drugs.[17] Reports are then submitted either to firms or directly to the FDA. If the firm receives the report, it is legally required to submit it to the FDA. However, the reports submitted by manufacturers still ultimately depend on physicians, clinicians, and others who submit them to the firm. All reports are then entered into the AERS database. Since the FDA is particularly interested in new risk information, it encourages the reporting of side effects that are unexpected (not reflected in the drug's labeling) and serious reactions such as those that are fatal, life-threatening, permanently disabling, or those requiring hospitalization.[18] In this way, AERS serves as an alert system for regulators about the serious unanticipated effects of drugs in the population that were not detected in clinical trials (Kessler, 1993).[19]

Although larger or longer clinical trials could reveal more information about drug safety (or efficacy) prior to approval, such studies are costly and may further delay approval. Furthermore, there will always be some uncertainty about drug safety at the time of approval. For this reason, regulators rely on post-marketing surveillance of ADRs after approval to learn more about drug risks in the population. Particularly in a drug's first few years on the market, the ADR reports received yield new information about drug risks. These reports often lead to improvements in product labels, new warnings, better guidelines for use (Kessler, 1993; Cross et al., 2002). ADR reports can also lead the FDA to withdraw a drug from the market.[20] However, it often takes years before such risk information is conveyed to physicians and the public (Lasser et al., 2002).

4.1.1. Pros and cons of the ADR data

The strengths of this database are that it covers all geographic regions, all patient populations (including understudied populations), and all marketed drugs. Strom (1994) suggests that the size and breadth of the database make it very useful for studying relatively rare drug reactions (not detected in clinical trials), drug reactions among infrequently used drugs (i.e. orphan drugs), and drug interactions.

The ADR data also have limitations. First, there is underreporting of ADRs because health professional participation, the primary source of reports for firms and the FDA, is voluntary. Second, no evidence of causation is required to submit an ADR report because these reports serve as the first alert to firms and regulators about possible new drug risks. This

[17] Baum et al. (1994).

[18] CDER (1996).

[19] Regulators encourage reporting of serious unanticipated ADRs after approval to learn more about drug risks associated with actual use. Data show that most reporting occurs for NCEs in their first 2 years on the market and serious unanticipated ADRs are more likely to be reported than ADRs for known, anticipated risks. However, certain drugs such as those with a black box warning for serious, known risks at the time of approval may also see greater reporting of ADRs.

[20] Graham et al. (2003) reports how Trogilitazone, approved in 1997, was withdrawn from the market in 2000 after 94 reported cases of liver failure and 65 deaths.

suggests that there will be some uncertainty about causation when a report is made. Third, the ADR database has no information on the level of drug utilization in the population. It only provides information on those who report ADRs. Fourth, other factors, such as the length of time on the market, whether or not a drug was launched abroad, the conditions treated, disease characteristics, and patient characteristics, may also influence ADR counts. These limitations will be addressed in the empirical analysis as follows.

4.1.2. Addressing the data limitations

Underreporting is addressed in three ways. First, the analysis focuses on the set of drugs (new chemical entities) and the marketing interval (first 2 years on the market) for each drug during which health professional incentives to report ADRs are the highest. New chemical entities (NCEs) are new molecular compounds not previously approved in the United States, excluding biologics, vaccines, diagnostic agents, and new salts, esters, and dosage forms of previously approved compounds (see Kaitin et al., 1994; Kaitin and Manocchia, 1997; Kaitin and Healy, 2000; Kaitin and Cairns, 2003 for the complete list of NCE approvals).[21] Uncertainty about drug safety tends to be the greatest for these innovative drug products because they are chemically new. ADRs are reported more frequently for new innovative drugs than for older drugs (Carson et al., 1994). Data show that ADR reporting peaks at the end of a drug's second year on the market and then declines thereafter (CDER, 1996). For this reason, the dependent variable in this study will consist of the ADRs reported in a drug's first 2 years after FDA approval.[22] Using this 2-year interval reduces any reporting bias that may arise from differences in the length of time that drugs have been on the market. Second, the analysis focuses on ADRs that are the most serious and hence least likely to be underreported. Third, the analysis will include year-specific effects to control for trends in underreporting or reporting over time.

To address uncertainty about causation, the analysis will focus on ADRs in which the individual filing the report identified the drug as the primary one suspected of causing the reaction.[23] Suspect status represents information about the reporter's or health professional's belief about causation. This information is contained in the ADR report by a category in which the reporter indicates the name of the primary suspect drug.

To address the lack of drug exposure data, drug utilization data from the Medical Expenditure Panel Surveys (MEPSs) from 1996 to 2003 is included in the analysis. The MEPS database (including the prescription drug survey) is designed to be nationally representative of the non-institutionalized civilian population in the US.[24] The prescription drug surveys were used to determine a proxy for drug utilization, that is the number of prescriptions among the drugs studied.[25] Possible limitations of the MEPS database will be discussed in the next section and addressed in the analysis.

To address the other influences on ADR reporting, the analysis will include several variables to control for the impact of the conditions being treated, disease characteristics, ADR patient characteristics, and year-specific effects. Building on prior research, the analysis will also include new controls for the effects of black box warnings at the time of approval as a factor that may influence ADR reporting, whether a drug was launched abroad prior to its US approval, and the length of (the US launch) lag to test for informational spillovers from abroad. More discussion of these variables is contained in the next section.

[21] Biologics are not included in this analysis. Unlike drugs, biologics are made from living sources such as proteins whereas drugs are chemically synthesized. Biologics were also typically approved the Center for Biologic Evaluation and Research (CBER), not in center that approves new drugs, CDER.

[22] New drugs are not marketed on the date of FDA approval, but typically at a later date referred to as the date of drug launch. My ADR counts reflect data from the first two calendar years after a drug's approval year to account for the lag in launching a drug following approval. No adjustments are made for the month of drug approval. Reports received by the agency in those years are included in the drug's count.

[23] Use of other drugs listed on the ADR report such as concomitant or secondary suspect drugs, could add noise to this measure.

[24] MEPS prescription drug data improves upon previous household health surveys by supplementing the household response data with data from pharmacy providers. MEPS obtains computerized printouts of prescribed medicines from pharmacy providers and uses written data forms from pharmacy providers when computerized printouts are unavailable. This information has not been previously used in household health surveys. MEPS also encourages improved reporting of drug utilization by asking household respondents about drugs prescribed in conjunction with other medical events, such as hospital stays, ER visits, and doctors visits (Moeller et al., 2001). Because patients who are institutionalized are not included in the survey, the survey is not likely to measure the use of new NCEs in that setting.

[25] Each year MEPS expenditure and use prescribed medicines totals are benchmarked to several outside sources with similar drug data (e.g., MCBS, IMS, NHA) to make sure that MEPS estimates are within reason of similar estimates produced by other sources. Each year MEPS estimates are compared to previous years MEPS estimates to ensure the estimates are in general consistent with trends we expect to see and have seen in the past with the data. For more information as well as references to numerous articles that have used MEPS data, see http://www.meps.ahrq.gov/mepsweb.

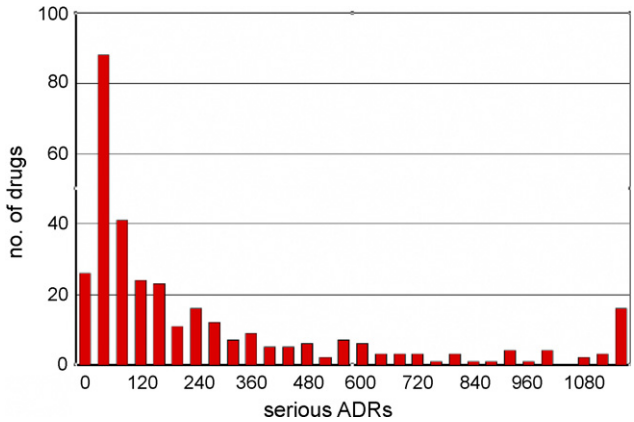

Fig. 2. Frequency distribution. Serious ADR counts: 1990–2001 NCEs.

4.2. The model

The dependent variables in this model are counts of ADRs for each drug i in the first 2 years following its FDA approval year t.[26] Each ADR identifying a drug as the primary suspect drug is included in the count:

$$Y_i = \sum_{j=1}^{2} \mathrm{ADR}_{i,t+j}$$

Three different ADR counts are examined: serious ADR count, ADR hospitalization count, and ADR death count. The count of serious ADRs will include all ADRs with outcomes that were life-threatening, required hospitalization, required intervention to prevent permanent damage, resulted in disability, or resulted in death. The second count includes all ADRs that required the patient to be hospitalized. The third count includes all ADRs that resulted in the patient's death. The three counts represent differences in the severity of health consequences due to ADRs.

Given the dependent variables are discrete, non-negative integers, methods appropriate for estimating count data, such as Poisson regression, should be used. Fig. 2 shows, however, that the frequency distributions for the ADR count data are highly skewed.[27] The data are over dispersed meaning that the conditional variance exceeds the conditional mean, which violates the Poisson assumption. Since over dispersion leads to downward bias in the standard errors from Poisson regression, the ADR data will be analyzed using negative binomial regression. The negative binomial regression models the variance of the distribution, σ^2 as a quadratic function of the mean, $E[Y_i] = \mu$, so that $\sigma^2 = \mu + \alpha\mu^2$ and where α is the dispersion parameter (Cameron and Trivedi, 1998).[28]

The empirical analysis models variation in the ADR count Y_i for drug i as a function of the drug's FDA review time, the user fee reform, PDUFA, as well as other important covariates including drug characteristics $X_{i,t}$, the conditions treated $C_{i,t}$, ADR patient characteristics $P_{i,t}$, and year-specific effects:

$$E[Y_i] = \exp(\beta_1 \text{ review time}_i + \beta_2 \,\mathrm{PDUFA}_i + \psi X_{i,t} + \phi C_{i,t} + \delta P_{i,t} + \gamma_t + \epsilon_{i,t}) \tag{1}$$

The variable review time$_i$ is defined as the time (in months) between the date of first submission of a new drug application and the date of FDA approval. This period includes the time that regulators spend reviewing the application and the time that firms take to respond to regulator requests for additional information to support the application. This interval is the key performance measure that Congress (and the industry) have used to evaluate agency performance.

[26] The calendar year of FDA approval is used as is the calendar year of the date of receipt of the ADR report. All data including counts and other variables including ADRs are coded by calendar year only.

[27] The frequency distributions for the other ADR counts are similar. Since 2-year cumulative counts are used, most drugs have positive counts. Only 7% of the drugs have zero serious ADR counts, 10% of the drugs have zero ADR hospitalization counts, and 20% have zero ADR death counts.

[28] A zero-inflated Poisson (ZIP) model was also considered for ADR deaths since there was a higher frequency of zero counts, but the log likelihood statistics indicated that it provided a poor fit relative to the negative binomial (NB) model (LL = −8039 for the ZIP model and LL = −1219 for the NB model).

The coefficient for this variable β_1 will measure the extent to which the speed of new drug review is associated with a drug's ADR count. Hypothesis 1 predicts that a negative sign for this coefficient $\beta_1 < 0$, which suggests that faster drug reviews are associated with higher counts of adverse drug reactions or increased drug-related risks.[29]

To measure the effects of PDUFA beyond review speed, two alternative PDUFA variables are considered. PDUFA(approvals), equal to 1 for all drugs approved on October 29 1992 or later, will measure the effect of the reform on all drugs approved in the PDUFA era. This PDUFA measure will include some drugs that were submitted prior to PDUFA, but approved after the reform, referred to as the backlog of unapproved drugs. Since there was a PDUFA performance goal to eliminate the backlog, it is reasonable to examine the impact of PDUFA pressures on these drugs along with the drugs submitted and approved in the PDUFA era.[30] A second variable PDUFA(submissions), equal to 1 for drugs submissions in the PDUFA era, is also considered to examine the sensitivity of the results to this definition of PDUFA. PDUFA(submissions) does not contain drugs from the backlog, but will only measure the effects of the reform on applications submitted and approved in the user fee era. The two variables will reveal the extent to which PDUFA pressures affected these two sets of drugs (PDUFA submissions and all PDUFA approvals).[31] Hypothesis 2 predicts a positive coefficient for $\beta_2 > 0$. This coefficient will measure the extent to which drug-related risks have increased among these two sets of drugs. If PDUFA primarily affects new drug safety through the speed of new drug review with no residual effect, then β_2 will not differ from zero. However, if other PDUFA-related factors, besides the speed of review, contribute to increased drug-related risks, then coefficient for the PDUFA will be positive.[32]

To examine Hypothesis 3, the interaction term review time × PDUFA will be added to the model. The coefficient for the interacted term will reveal whether the effect of drug review speed on drug-related risks significantly changed in the user fee era. A positive coefficient for this interaction term will indicate that the speed of review has a larger effect on drug-related risks in the user fee era. A negative coefficient could indicate that user fees helped regulators reduce the effects of review times on drug risks. To examine Hypothesis 4, the interaction term novel drug × PDUFA will be added to the model, where novel drug is a dummy variable indicating a therapeutic novelty drug. The coefficient for this interacted term will indicate whether the risks associated with novel drugs significantly changed after the user fee reform. Hypothesis 4 predicts a positive coefficient for this interaction term, which suggests that drug-related risks among novel drugs increased in the PDUFA era.

Drug characteristics $X_{i,t}$ will also influence drug-related ADR counts. Since drugs with higher utilization may be expected to generate more ADRs than drugs with lower utilization, a variable for utilization is included, log(utilization), equal to the logarithm of the number of prescriptions associated with each drug 2 years after approval. The coefficient for this variable (an elasticity) is constrained to be the same for all drugs. Data from the 1996–2003 Medical Expenditure Panel Surveys (MEPSs) prescription drug files are used to estimate drug utilization. Since orphan diseases affect small patient populations, orphan drugs may not be well represented in the MEPS data. The variable, orphan, defined as 1 for drugs that received an orphan designation from the FDA and 0 otherwise, is included to test for a discontinuity in the way that orphan drugs relate to ADRs. A negative coefficient for orphan could be indicative of the low utilization of orphan drugs as a group relative to other drugs instead of providing an indication of below average risks.[33] A positive coefficient for orphan could be indicative of drugs that have above average risks relative to other drugs.

A second important drug characteristic is a drug's priority rating, which will control for the variation in ADRs among therapeutically novel and less novel drugs. The dummy variable novel drug is equal to 1 for all drugs receiving a priority review rating from the FDA and 0 for drugs receiving a standard rating from the FDA.[34] The FDA gives a

[29] Alternative analyses using a logged value of review time will also be examined to test the sensitivity of this functional form.

[30] See http://www.fda.gov/ope/pdufa/report95/appenda.html for the statement of this performance goal.

[31] On the merits of approval year versus submission year measures, see Olson (2004c) and Carpenter (2004) for competing viewpoints on this issue.

[32] An alternative approach to measuring the effects of PDUFA is to use a trend variable. An alternative estimation will examine the effect of the variable, FeeTrend, which is equal to 1 for drugs approved from October 29 1992 to December 31 1993, 2 for drugs approved in 1994, 3 for drugs approved in 1995, 4 for drugs approved in 1996, 5 for 1997 approvals, and 6 for 1998 approvals and 0 otherwise.

[33] Orphan drug status is designated for diseases that affect fewer than 200,000 patients. Since some orphan diseases like ALS affect only 1–2 per 100,000, the use of such drugs may not be reflected in even large health surveys like the MEPS. Only 10 of the 62 orphan drugs are included in the MEPS data in the 2 years after the drug's approval.

[34] Before the P, S rating system was adopted in 1992, the FDA used the A, B, C, AA, system. Drugs rated A, B, and AA (for AIDs drugs) were defined as having some therapeutic advance over existing remedies. For this reason, drug having an A, B, and AA rating are assumed to have a

priority rating at the time of submission to any new drug believed to offer significant therapeutic benefits over existing remedies, or representing the first of their kind in a therapeutic class.[35] Novel drugs may be associated with increased risk (Olson, 2004b) for two reasons. First such drugs are considered more urgent by regulators and so they may approve them based on less safety data.[36] Second, FDA has tried to approve these drugs more quickly than less novel drugs over time. Including the novel drug variable in the analysis will reduce the extent to which between-group differences (among priority and standard rated drugs) influences the coefficient for review time. Hence, results will reflect the within class variation among novel and less novel drugs and not the between class variation.[37]

In addition to a drug's priority rating, the variable accel approval is equal to 1 for drugs designated as accelerated approvals or subpart E applications. Subpart E consists of procedures to help facilitate the development and submission of NDAs for life-threatening or debilitating illnesses. Accelerated approvals, which target the same patient populations, are permitted to use surrogate endpoints; i.e. endpoints other than survival or morbidity. These drugs are expected to be more risky for patients because they have less clinical data about drug effects and drug safety at the time of approval.[38]

The variable injectable equal to 1 for drugs requiring injection is included as a third drug characteristic in the analysis. Injectable drugs tend to be directly administered in hospitals and physician offices and not by prescription, which could lead to increased reporting of adverse effects for these drugs.[39]

Whether a drug had a black box warning at the time of approval is a fourth drug characteristic that may influence ADR counts. Black box warnings on drug labels and communications are used to inform patients about serious known risks (i.e. increased risk of stoke). Since drugs with serious known risks may have more anticipated reporting, it is important to control for whether a drug had a black box warning at the time of approval. The variable black box, equal to 1 for drugs that had a black box warning with such risk information at the time of approval, will control for any anticipated increased in ADR reports for these drugs.[40] A black box may also indicate whether there are restrictions for specific patient groups. Eleven of the drugs receiving a black box at the time of approval had a single warning that the drug was contraindicated for pregnant women. These drugs are distinguished from those above by a separate variable preg BBox.

Two variables are added to control for the effects of foreign drug launches on ADR counts. The variable foreign launch, which is equal to 1 for drugs first launched abroad prior to US approval, is included to control for variation in ADRs among drugs first used abroad and those first approved in the US. Potential drug risks could affect firms' drug launch decisions. Firms could use foreign first drug launches as a way to learn more about risky drugs prior to US approval.[41] Consequently, such drugs may generate more ADRs upon US launch than those that are first approved in the US. The variable launch lag, which is equal to the year of FDA approval minus the year of first drug launch abroad, is included to control for the effects of any informational spillovers (safety information that comes from abroad prior to FDA approval) on ADR counts.

Differences in the conditions treated $C_{i,t}$ may also affect drug-related ADR counts. Patients with certain conditions may be at greater risk of having an ADR. For example, patients with AIDS/HIV may also be more likely to experience an ADR. By including variables for the conditions being treated, the analysis will control for the variation among ADRs due to condition-specific effects. The more cohesive the grouping of conditions by variable, the more control offered. The condition-specific variables include hypertension, high cholesterol, other heart disease, bacterial infection, AIDS/HIV, other infection, cancer, inflammation, arthritis, anesthetics (for surgery), other pain relief, asthma/allergy/respiratory,

priority designation. Drugs given a C rating were defined as offering no therapeutic gain over existing remedies, which is how S rated drugs are defined.

[35] Drugs receiving standard ratings, while chemically new, are not believed to offer important advances over existing remedies, and hence not considered as urgent.

[36] Several AIDS drugs were approved in the 1990s with less than complete safety and efficacy data.

[37] Review deadlines differ only by the priority and standard drug designations.

[38] These submissions are also given P or S ratings, which determine their drug review deadlines. Since many, but not all of these drugs, are deemed novel by regulators, a positive coefficient for this variable will reveal the extent to which the risks associated with accelerated approvals differ significantly from the risks associated with novel drugs.

[39] Alternatively, injectables may be less likely to be represented in the MEPS prescription drug survey and hence the variable may also account for unmeasured utilization effects.

[40] Since it takes time to analyze ADRs and learn about new side effects in the population, new boxed warnings typically will not be seen until several years after approval (see Lasser et al., 2002). Agency negotiations with firms about such labeling changes after approval adds further delay.

[41] Alternatively, regulators may spend more time on risky drug applications, which could also delay their US approval.

mental health diseases, epilepsy, diabetes, other endocrine disorders, gastrointestinal illness, Parkinsons/MS, and glaucoma.[42] Disease classifications were made based on the information from a drug's label and approved indication and then confirmed by an online medical sources (i.e. WebMD).[43] A separate drug effect is included for Viagra, which is used to treat impotence, because it was identified ex ante as an extreme outlier. This drug had the highest number of ADR deaths among all drugs studied in their first 2 years on the market (including those withdrawn).[44] The analysis will also be conducted excluding Viagra to determine the sensitivity of the results to its inclusion in the study.

The analysis includes two variables to control for chronic and fatal diseases. The variable chronic, equals 1 for diseases that are chronic in nature and hence require long-term treatment.[45] The variable fatal equals 1 for diseases in which 50% of the diagnosed patients are expected to die within 5 years at the time of a drug's approval. This measure relates to the effects of a disease, not the effects of the target drug. Chronic and fatal are disease characteristics that may influence the likelihood of experiencing an ADR. Patients who have chronic or fatal diseases may have a greater chance of having an adverse effect because they are in poor health. The predicted coefficients for both variables are positive.[46]

Characteristics of the patients $P_{i,t}$ reporting ADRs may also affect count levels. The variable age is defined as the average age of the patients who experience ADRs. Age may proxy for the health status of the patient who experiences an ADR. If older patients take more medications, they may also have a greater chance of experiencing ADRs. Both rationales suggest that age will be positively associated with ADR counts. A second variable female is defined as the percentage of females represented in an ADR count. This variable controls for gender differences in the likelihood of experiencing or reporting ADRs and its predicted coefficient is positive. Although these variables are not ex ante measures of heterogeneity in population risk by NCE, they do control for ex post heterogeneity in the age and gender of mix of patients reflected in the ADR counts. The coefficients will reveal the extent to which the age and gender of those who experience ADRs is systematically related to the count.

Finally, approval year effects γ_t are included in the model to control for reporting or underreporting trends common to drug approvals in that year. Year effects would also pick up the effects of idiosyncratic changes in a given year such as the introduction of Medwatch or other significant events that may affect reporting trends for drug approvals that year.

An empirical issue to address is the possible endogeneity of the review time variable. Drugs believed to have high risks ex ante may result in regulators taking more time to review those drugs rather than, as suggested in this paper, that faster reviews lead to increases in ex post drug risks. This would result in a positive coefficient for review time. Alternatively, drugs that are risky may also have the greatest therapeutic benefit, and hence regulators may try to accelerate the approval of these drugs. In fact, regulators introduced the priority rating exactly for this purpose. The novel drug variable, which reflects a priority rating, will control for a regulator's ex ante assessment of drugs offering the greatest therapeutic benefits. As discussed earlier, this variable will remove the influence of between group (priority and standard) variation on the estimated coefficient for review time.

To investigate the possible endogeneity of review time, an exogeneity test for non-linear models (Wooldridge, 2002), which uses an omitted variables approach, is performed. The test requires instruments for review time be developed. Firm characteristics are chosen as instruments because they are correlated with the speed of review Olson (1997), but not correlated with post-approval drug risks.[47] Instruments include firm research intensity (rd/total sales), firm specialization in pharmaceuticals (drug sales/total sales), firm size (log(employees)), current applications under review, applications approved since 1990, an indicator for a US firm, an indicator for a new firm (no prior sales). Data

[42] The disease classification system is similar to that used in Olson (2004a, b, c) with two refinements, a category for MS/Parkinsons and a category for glaucoma. Additional approvals in these categories made these refinements feasible and offered better control than more aggregate categories.

[43] The omitted case is an 'other' category, which consists of drugs that had small cells and did not fit into existing classifications. Other includes 22 drugs representing insomnia (4), enzyme deficiency (2), severe kidney disease (3), poisoning remedies (2), weight loss (2), stimulant for excessive sleepiness (1), overactive bladder (1), psoriasis (1), a metabolic disorder (1), SCID (1), a drug to reduce pupil size (1), Sjogren's Syndrome (1), a chemical warfare agent (1), and a drug for age spots (1).

[44] Prescription drug utilization data from the MEPS or other databases focusing on retail outlets may understate Viagra use because of illegal nonprescription distribution channels.

[45] The characterization of chronic is based on the disease for which a drug is targeted.

[46] The chronic and fatal variables allow for additional cross-sectional variation that may cut across the conditions described above.

[47] For instance Merck is an innovative, R&D intensive, specialized pharmaceutical firm with a strong scientific reputation. Such characteristics could not predict the safety problems with Merck's drug Vioxx and its subsequent withdrawal from the market.

for firm characteristics were obtained from the annual reports of firms who had NCE approvals in the year prior to drug approval. The first stage also included variables for drug novelty and the other exogenous variables in (1).[48] Review time was estimated using linear regression.[49] The first stage regression residuals were then added to the negative binomial regression of ADR counts. The coefficient for the first stage residuals was then tested against zero. For all ADR counts, the test did not provide evidence against the null (exogeneity of review time).[50]

4.3. Data

The drug names, therapeutic novelty ratings, the orphan drug status, and the review times for the 333 new chemical entities approved in 1990–1992, 1993–1995, 1996–1998, and 1999–2001 are published in Kaitin et al. (1994), Kaitin and Manocchia (1997), Kaitin and Healy (2000), and Kaitin and Cairns (2003), respectively. The form of an NCE is listed on the FDA's Drugs@FDA website for its given approval date. If the drug is listed as an injectable, then the variable was set equal to 1, otherwise the variable is zero.[51] The country and year of first drug launch were obtained from PharmaProjects.[52] Six drugs did not remain in the market for two full years after approval and so are excluded from the analysis because their ADR counts may be biased.[53] An additional 17 drugs did not have foreign launch data. Hence the analysis is performed on a total of 310 new drug approvals, for which foreign launch data were available.

The ADRs come from the FDA's Spontaneous Reporting System (SRS) and its Adverse Event Reporting System (AERS). Beginning with the ADR reports received after October 1997, the FDA changed the name of the ADR database from the SRS to AERS. Similar information is contained in each database although in the AERS, FDA requests that primary and secondary suspect drugs be listed.[54] Data collected from each report include clinical information on the adverse drug reaction, the date of the reaction, whether the drug is suspected of causing an ADR (indicated by an S in the data prior to October 1997 and PS, primary suspect, after October 1997), the patient's age, and the patient's gender, and the source of the report. ADRs that cite a particular drug can be attributed to the branded drug since generic versions of these drugs are not available until after patent expiration. Most ADRs cite the drug's brand name. Reports from foreign sources (those outside the US) are excluded from the analysis. Less serious ADRs and ADRs that fail to identify a suspect drug are not included in this analysis. Black box warning data at the time of approval was obtained from the Physicians' Desk Reference (in the year of the drug's first listing) and from the FDA's website.

Prescription drug utilization data from the 1996 to 2003 Medical Expenditure Panel Surveys are also used. Data from the 1996 prescription drug survey are used to proxy for utilization among the 1990–1993 drug approvals (ADRs from 1991 to 1995).[55] For each of the drug approvals in 1994, 1995, 1996, 1997, 1998, 1999, 2000, and 2001, utilization data are taken from 1996, 1997, 1998, 1999, 2000, 2001, 2002, and 2003 prescription drug surveys, respectively.

The variables used to control for the conditions treated are constructed from each drug's FDA-approved indication. This information is contained in the FDA's drug approval files. The diseases were coded as chronic if the disease required on-going or long-term drug treatment based on reviews of medical literature. Diseases were classified as fatal if the 5-year survival rate was less than or equal to 50%. Survival rate data were sought prior to a drug's approval to avoid the prospect of any possible influence of the drug on survival.[56]

[48] The instruments were interacted with the PDUFA variable since this reform altered the effect of these determinants on drug review times (see Olson, 2004a).

[49] The first stage regression had an adjusted R^2 of 0.75. Coefficients for the instruments from the first stage regression and the F-statistic on the excluded instruments are reported in Appendix A.

[50] The coefficients for the first stage residuals were not significantly different from zero in any ADR negative binomial regression.

[51] The analysis does not consider or control for supplemental NDAs, which may represent a different formulation of an NCE.

[52] I thank Margaret Kyle for making the foreign launch data available to me for this analysis.

[53] Four of the drugs withdrawn from the market list no reported serious ADRs in the AERS database in the 2 years after approval. Since the drugs were withdrawn and given lags in data entry, ADRs may have never been inputted for these drugs.

[54] The addition of secondary suspect drugs is new to AERS. Prior to that FDA only requested the name of the drug suspected of causing the adverse event.

[55] To determine whether the results are sensitive to this assumption, the analysis was conducted for the 1994–1998 approvals, which excluded the 1990–1993 approvals. The coefficients for review time remained negative and statistically significant in all regressions. Other results remained similar.

[56] National Cancer Institute's SEER database, 1985–1989 data, was used to code fatal for different cancers. Fatal determinations were the same under the 1993–1997 SEER data. Fatal determinations for the remaining diseases were based on major journal articles (from JAMA, NEJM) in

5. Results

5.1. Summary statistics

Table 1 reports the summary statistics for the variables used in the analysis. The summary statistics were calculated for the 310 new chemical entities approved in 1990–2001 that remained in the market for two full years after their approval for which foreign launch data was available. The 310 drugs have a total of 96,751 serious ADRs, 57,511 ADRs that required hospitalization, and 17,797 ADRs that resulted in death in the 2 years after FDA approval.

Among the 1990–2001 approvals, the average count of serious ADRs is 312 in the 2 years after FDA approval. The average number of ADR hospitalizations is 186 and the average number of ADR deaths is 57 among this set of drugs in their first 2 years on the market. The standard deviations indicate substantial variation within each ADR category, which is also evident in Fig. 2.

The mean FDA review time among these drugs is 21 months. There is a pattern of declining average review times in the PDUFA era. For new drugs approved in 1990–1992 prior to PDUFA, the mean review time was 31 months. For new drugs approved in 1993–1995 in the first 3 years after PDUFA, the mean review time fell to 24 months. For new drugs approved in 1996–1998, the mean review time fell to 16.8 months. For new drugs approved in 1999–2001, the mean review time was 16.7 months. Eighty-one percent of the drugs were approved after the introduction of user fees. Sixty-nine percent of these drugs were submitted after PDUFA was signed into law. Forty-two percent of the drugs represent therapeutically novel drugs. Accelerated approvals account for 13% of the drugs. Seventeen percent of the drugs were given black box warnings at the time of approval, while another 4% had the pregnancy warning only in the black box. Fifty-seven percent of the drug approvals were first launched abroad prior to FDA approval. The percentage of drugs first launched abroad declines in the 1990s from 67% in 1990–1992 and 1993–1995 to 52% in 1996–2001. The average lag until US launch is 2.2 years, which also shows a declining trend over time. Twenty percent of the drugs are classified as orphan drugs affecting small patient populations.

The mean ages of patients who experience ADRs are in the mid- to late 1940s, but the standard deviations are large ranging from 21 years for serious ADRs to 28 years for ADRs resulting in death. The data also indicate that the average percentage of women reflected in the ADR counts is 45% among serious ADRs and ADR hospitalizations and 34% of ADR deaths per drug. The standard deviations show that there is substantial variation in the percentage of female ADR patients.

5.2. Regression results

Table 2 presents the results from the serious ADR count regressions. Two specifications are presented to examine the sensitivity of the results to the different PDUFA measures. Column (1) presents the results for the regression with PDUFA(approvals). Column (2) presents the results with PDUFA(submissions).

Results show that faster drug review times are associated with increased counts of serious ADRs. The coefficient for review time is negative and significant at the 0.05 level in both columns of Table 2 indicating that the result is not sensitive to the PDUFA measure used. Results show that a 1 month reduction in a drug's review time is associated with a 1.2–1.3% increase in the number of serious ADRs received in the first 2 years after FDA approval. A scatter plot of the adjusted counts versus review times using the results in column (1) is shown in Fig. 3. The figure depicts the inverse relationship between review speed and the adjusted counts.[57] Several elevated counts are noticeable at both the 6 months and at later months.[58] The results were not sensitive to the exclusion of these outliers from the analysis. The coefficient for review time remained negative and highly significant for all ADR counts.[59] Hence increased review speed has led to more serious drug reactions for patients, consistent with Hypothesis 1.

years preceding drug approval (i.e. 1990 data were used for AIDS). National Center for Health Statistics at the Centers for Disease Control was used for AIDs opportunistic infections. Information from an ALS organization was used to code ALS as fatal.

[57] Scatter plots for the predicted counts in Tables 3 and 4 showed a similar pattern.

[58] These drugs, Fosamax, Taxotere, Aricept, Celebrex, Viagra, Vioxx, Exelon, and Reminyl had predicted counts greater than 2S.D. above the mean predicted count.

[59] Similar results were found when the logged value of review time was used in the regression. The coefficient (S.E.) for logged review time in Table 2 was −0.26 (0.14) in column (1) and −0.26 (0.15) in column (2), both significant at the 0.1 level.

Table 1
Summary statistics: 1990–2001 drug approvals ($N = 310$) [a]

	Mean	S.D.
Dependent variables		
Serious: serious ADRs for drug *i* in the 2 years after FDA approval	312	662
Hospital: ADR hospitalizations for drug *i* in the 2 years after FDA approval	186	392
Death: ADR deaths for drug *i* in the 2 years after FDA approval	57	148
Independent variables		
Review time: the drug's FDA review time (in months)	21	19
PDUFA(approvals): equals 1 if drug approved after 29 October 1992	0.81	0.39
PDUFA(submissions): equals 1 if drug submitted after 29 October 1992	0.69	0.46
Black box: equals 1 if boxed warning at time of approval	0.17	0.38
Preg BBox: equals 1 if boxed warning with pregnancy contraindication only	0.04	0.20
Foreign launch: equals 1 if launched abroad prior to FDA approval	0.57	0.50
Launch lag: year of FDA approval − year of foreign launch abroad	2.20	3.93
Novel drug: equals 1 for priority-rated drugs by FDA (indicates novelty)	0.42	0.49
Accel approval: equals 1 for accelerated approvals or subpart E applications	0.13	0.34
log(utilization): logarithm of the number of prescriptions for each drug	2.45	2.41
Orphan: equals 1 for orphan drugs (patient populations less than 200,000)	0.20	0.40
Injectable: equals 1 for injectable drugs	0.24	0.43
Age (s): average age of the patients reflected in a drug's serious ADR count	48.0	21.0
Age (h): average age of the patients in a drug's ADR hospitalization count	47.0	22.0
Age (days): average age of the patients in a drug's ADR death count	43.0	28.0
Gender (s): the percentage of female patients in a drug's the serious ADR count	0.45	0.26
Gender (h): the percentage of female patients in ADR hospitalization count	0.45	0.28
Gender (days): the percentage of female patients in a drug's ADR death count	0.34	0.29
Patient conditions		
AIDS/HIV: equals 1 for drugs intended for AIDS/HIV	0.07	0.26
Bacterial infections: equals 1 for drugs intended for bacterial infections	0.07	0.26
Other infection: equals 1 for drugs intended for other infections	0.07	0.25
Cancer: equals 1 for drugs intended to treat various cancers	0.13	0.34
Hypertension: equals 1 for drugs intended to treat hypertension	0.08	0.27
High cholesterol: equals 1 for drugs intended to treat high cholesterol	0.02	0.15
Other heart disease: equals 1 for drugs intended for other heart diseases	0.06	0.25
Epilepsy: equals 1 for drugs intended to treat epilepsy	0.03	0.17
Inflammation: equals 1 for drugs intended to treat inflammation	0.02	0.14
Arthritis: equals 1 for drugs intended to treat arthritis	0.03	0.17
Anesthetics: equals 1 for anesthetics that may be used in surgery	0.03	0.16
Other pain relief: equals 1 for analgesics intended for other pain relief	0.04	0.20
Mental health: equals 1 for drugs intended for mental health conditions	0.05	0.21
Asthma/allergy/respiratory: equals 1 for drugs for asthma, allergy, and respiratory illnesses	0.08	0.27
Diabetes: equals 1 for drugs used to treat diabetes	0.04	0.19
Other endocrine disorders: equals 1 for drugs used to treat other endocrine diseases	0.06	0.23
Gastrointestinal disease: equals 1 for drugs used to treat gastrointestinal illnesses	0.02	0.14
Parkinsons/MS: equals 1 for drugs used to treat Parkinsons disease, MS, or ALS	0.02	0.15
Glaucoma: equals 1 for drugs used to treat glaucoma	0.03	0.16
Disease characteristics		
Chronic: equals 1 for drugs used to treat chronic conditions	0.48	0.50
Fatal: equals 1 for drugs used to treat fatal conditions	0.11	0.31

[a] This total excludes 17 drugs for which foreign launch data were unavailable and the 6 drugs that were withdrawn from the market in the 2 years after their approval. They are Manoplax, Omniflox, Duract, Raxar, Posicor, and Lotronex.

Table 2
Negative binomial regression results for serious ADR counts: 1990–2001 NCEs

Variable	(1)	(2)
Review time	−0.012** (0.005)	−0.013** (0.005)
PDUFA(approvals)	−0.08 (0.55)	–
PDUFA(submissions)	–	−0.09 (0.43)
Black box	0.74*** (0.23)	0.74*** (0.23)
Preg BBox	−0.51 (0.40)	−0.51 (0.40)
Foreign launch	0.57*** (0.18)	0.57*** (0.18)
Launch lag	−0.06*** (0.02)	−0.06*** (0.02)
Novel drug	0.94*** (0.18)	0.94*** (0.18)
Accel approval	0.73*** (0.29)	0.72*** (0.29)
log(utilization)	0.19*** (0.04)	0.19*** (0.04)
Orphan	−0.25 (0.23)	−0.25 (0.22)
Injectable	0.06 (0.21)	0.05 (0.21)
Age	0.05*** (0.006)	0.05*** (0.006)
Gender	2.07*** (0.38)	2.06*** (0.37)
AIDS/HIV	1.80*** (0.49)	1.80*** (0.49)
Bacterial infection	1.34*** (0.38)	1.35*** (0.38)
Other infection	0.77** (0.39)	0.76** (0.39)
Cancer	0.73* (0.40)	0.74* (0.40)
Hypertension	0.49 (0.43)	0.49 (0.43)
High cholesterol	0.41 (0.55)	0.42 (0.54)
Other heart disease	0.45 (0.40)	0.45 (0.40)
Epilepsy	2.10*** (0.51)	2.08*** (0.52)
Inflammation	−1.06* (0.63)	−1.09* (0.65)
Arthritis	1.00** (0.50)	1.00** (0.50)
Anesthetics	1.34** (0.55)	1.36*** (0.55)
Other pain relief	0.30 (0.44)	0.29 (0.44)
Mental health	1.77*** (0.45)	1.76*** (0.45)
Asthma/allergy/respiratory	0.51 (0.40)	0.49 (0.41)
Diabetes	0.15 (0.48)	0.15 (0.48)
Other endocrine disorders	−0.18 (0.39)	−0.18 (0.39)
Gastrointestinal disease	−0.04 (0.59)	−0.05 (0.59)
Parkinsons/MS	0.84 (0.52)	0.84 (0.52)
Glaucoma	−2.14*** (0.53)	−2.14*** (0.53)
Chronic	0.73*** (0.23)	0.74*** (0.23)
Fatal	0.32 (0.29)	0.32 (0.29)
Viagra	3.99*** (1.16)	3.99*** (1.16)
Dum91	−0.02 (0.40)	−0.03 (0.40)
Dum92	0.57 (0.47)	0.53 (0.41)
Dum93	0.21 (0.69)	0.14 (0.43)
Dum94	0.49 (0.72)	0.46 (0.51)
Dum95	0.99 (0.72)	0.97* (0.53)
Dum96	1.14* (0.66)	1.14** (0.51)
Dum97	0.74 (0.66)	0.75 (0.54)
Dum98	0.96 (0.67)	0.97* (0.56)
Dum99	0.63 (0.68)	0.63 (0.53)
Dum00	0.61 (0.68)	0.62 (0.58)
Dum01	0.84 (0.67)	0.85 (0.56)
Constant	−1.37*** (0.54)	−1.34** (0.56)
Dispersion	1.16	1.15

Dependent variable = serious ADR count; $N = 310$. Results with PDUFA(approvals) in column (1). Results with PDUFA(submissions) in column (2). Standard errors are in parentheses.*Significant at the 0.1 level.**Significant at the 0.05 level.***Significant at the 0.01 level.

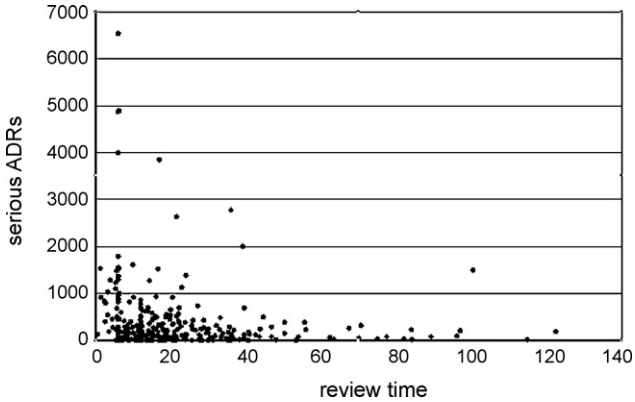

Fig. 3. Adjusted counts vs. review times.

Results show little additional impact of user fees on ADR counts after controlling for the effects review speed. Neither of the coefficients for PDUFA(approvals) in (1) or PDUFA(submissions) in (2) is significantly different from zero.[60] This suggests that the primary impact of user fees on ADRs occurs via the speed of review.[61]

In addition to review speed, black box warnings, foreign drug launch prior to US approval, and shorter US launch lags are associated with increased counts of serious ADRs. The coefficient for black box is positive and significant at the 0.01 level in (1) and (2), respectively, indicating that the drugs with such warnings also have higher counts of serious ADRs. The coefficient for the pregnancy boxed warning, although negative, is not significantly different from zero. The coefficient for foreign launch is positive and significant at the 0.01 level in (1) and (2), indicating that the drugs first launched abroad are associated with higher counts of serious ADRs when launched in the US. The length of the lag until US launch is negative as predicted, and significant at the 0.01 level in each column, which suggests that shorter lags have led to fewer informational spillovers (safety information from abroad), and consequently increased ADRs. After controlling for these factors, however, there is still an inverse relationship between the speed of FDA review and counts of serious ADRs.

The remaining coefficients show the impact of other influences on adverse drug reactions. The coefficient for log(utilization) is positive and significant at the 0.01 level in each column, which suggests that drugs taken by more people generate a higher number of serious ADRs. The coefficient for orphan drug is negative but not significantly different from zero. The coefficient for novel drug is positive and significant at the 0.01 level in each column, which suggests that drugs receiving a priority rating (indicating novelty) are associated with higher counts of serious ADRs than less novel drugs. In addition, the coefficient for accel approval is positive and significant, indicating that the drugs submitted as accelerated approvals or subpart E applications are associated with increased serious ADRs. Both age and female variables are positively related to serious ADRs in both columns, which indicates that the ex post heterogeneity in ADR patient age and gender is systematically related to the count of serious ADRs. The results suggest that the higher the mean age of patients (or percentage female) reflected in the count, the greater the number of serious ADRs.

Among the condition-specific variables, the drugs used for AIDS, anesthetics, arthritis, bacterial infections, other infections, cancer, epilepsy, and mental health conditions are all associated with higher counts of serious ADRs in each column of Table 2. Drugs used to treat AIDS, epilepsy, and mental health conditions are associated with the largest increases in serious ADR counts.[62] The only other coefficient that is larger is the one for Viagra. This coefficient implies that Viagra use is significantly associated with higher counts of serious ADRs. In contrast, drugs used to treat glaucoma are associated with smaller counts of serious ADRs. The significance of several of condition-specific effects suggests that patient conditions are important factors explaining variation in serious ADR counts. In addition, the coefficient for

[60] When backlog is included as a separate variable, its coefficient is also not significantly different than zero. Further, the coefficient for backlog is not significantly different from the coefficient for PDUFA(approvals), suggesting that the two sets of PDUFA approvals can be pooled.

[61] An alternative estimation using the FeeTrend variable instead of PDUFA produced a coefficient (S.E.) equal to 0.10 (0.56).

[62] Since the MEPS focuses on the non-institutionalized population, it is possible that utilization of new mental health NCEs from those in such institutions would not be in the survey. Then, the positive coefficient for mental health could reflect unmeasured utilization of mental health drugs. However, whether institutionalized patients are receiving NCEs in their first 2 years on the market is another relevant question.

chronic disease is positive and significant in (1) and (2), which suggests that the drugs used to treat chronic illnesses are associated with higher counts of serious ADRs. The coefficients for fatal conditions are positive but not significant in either column.

Table 3 shows that faster reviews are also associated with increased ADR hospitalization counts. The coefficient for drug review time is negative and significant at the 0.05 level in (1) and (2) indicating that the result is not sensitive to the PDUFA measure used. Results show that a 1 month reduction in a drug's review time is associated with a 1.1% increase in ADR hospitalizations in the first 2 years after FDA approval.[63] Hence increased review speed has also led to more adverse drug reactions that require hospitalization, which further supports Hypothesis 1.

Results show little additional impact of user fees on ADR hospitalizations after controlling for the effects review speed. As in Table 2, neither coefficient for PDUFA(approvals) or PDUFA(submissions) is significantly different from zero.[64] Hence, user fees appear to have little direct influence on ADR hospitalizations beyond what is captured by the speed of review.

As in Table 2, black box warnings, foreign launches prior to FDA approval, and shorter US launch lags are associated with increased counts of ADR hospitalizations in Table 3. The coefficient for black box is positive and significant at the 0.01 level in (1) and (2), further indicating that the drugs with such warnings also have more ADR hospitalizations. Foreign launch is also positive and significant at the 0.01 level in (1) and (2), further indicating that the drugs first launched abroad are associated with higher counts of ADR hospitalizations when launched in the US. The length of the lag until US launch is negative and significant at the 0.01 level, which further supports the presence of informational spillovers. Like Table 2, however, faster reviews are associated with increased drug risks even after controlling for all these influences on ADR counts.

The results for the drug utilization variables, drug novelty, and the characteristics of patients who reported ADRs are similar to those reported in Table 2. Drug utilization, drug novelty, the age and gender (percent female) of ADR patients are all significantly associated with increased ADR hospitalization counts. The coefficient for accelerated approvals, however, is not significantly associated with ADR hospitalizations. The condition-specific effects are also similar to those in Table 2. Drugs used to treat AIDS, epilepsy, and mental health conditions are still associated with the largest increases in ADR hospitalizations. Drugs used to treat chronic illnesses are associated with significantly higher ADR hospitalization counts in both (1) and (2). Once again the variable for Viagra has the largest coefficient, which suggests that Viagra use is significantly associated with increased counts of ADR hospitalizations.

Table 4 reports the results from the ADR death count regressions. As in the other two tables increased review speed was associated with higher ADR deaths counts. Significant coefficients for review time are observed in both columns of Table 4 indicating that the result is not sensitive to the PDUFA measure used. The coefficient indicates that a 1 month reduction in a drug's review time is associated with a 1.1–1.2% increase in a drug's ADR death count.[65]

Neither PDUFA measure is significant in Table 4. After controlling for review speed, PDUFA approvals and PDUFA submissions are not significantly associated with ADR death counts.[66] These results suggest that it was primarily through the speed of review that PDUFA affected ADR deaths. Novel drugs, accelerated approvals, drugs first launched abroad, and drugs with shorter US launch lags are associated with increased ADR deaths. The coefficient black box is positive and marginally significant, indicating that drugs receiving black box warnings are associated with more ADR deaths. In contrast to the other tables, the coefficient for preg BBox is negative and significant at the 0.05 level, indicating that contraindications relating to pregnancy in a boxed warning may have a protective effect on ADR deaths.

The drugs used to treat chronic illnesses and fatal illnesses are also associated with increased ADR death counts. The impact of the characteristics of patients who reported ADRs are similar to those in Tables 2 and 3. Several condition-specific effects are similar to those reported in Table 3. Some differences are that drugs used for hypertension are associated with significantly higher ADR death counts; drugs used for inflammation are associated with fewer ADR deaths; and the drugs used for cancer are no longer related to ADR deaths. Drugs used to treat bacterial infections,

[63] Similar results were found when the logged value of review time was used in the regression. The coefficient (S.E.) for logged review time in Table 3 was −0.30 (0.13) in column (1) and −0.31 (0.15) in column (2), both significant at the 0.05 level.

[64] Similar results are found when the alternative FeeTrend variable is used instead of PDUFA. The coefficient (S.E.) for FeeTrend is equal to 0.25 (0.53).

[65] When logged review time was used in Table 4, the coefficient (S.E.) was −0.16 (0.15) in column (1) and −0.14 (0.17) in column 2, neither of which are significant.

[66] Results from the alternative estimation including FeeTrend produced similar results. The coefficient (S.E.) for FeeTrend was −0.39 (0.64).

Table 3
Negative binomial regression results for ADR hospitalization counts: 1990–2001 NCEs

Variable	(1)	(2)
Review time	−0.011**(0.005)	−0.011**(0.005)
PDUFA(approvals)	0.10 (0.52)	–
PDUFA(submissions)	–	−0.06 (0.42)
Black box	0.64*** (0.22)	0.64*** (0.22)
Preg BBox	−0.52 (0.39)	−0.52 (0.39)
Foreign launch	0.55*** (0.17)	0.55*** (0.17)
Launch lag	−0.06*** (0.02)	−0.06*** (0.02)
Novel drug	0.96*** (0.18)	0.96*** (0.18)
Accel approval	0.43 (0.29)	0.43 (0.29)
log(utilization)	0.23*** (0.04)	0.23*** (0.04)
Orphan	−0.24 (0.22)	−0.25 (0.22)
Injectable	0.03 (0.21)	0.03 (0.21)
Age	0.05*** (0.006)	0.05*** (0.006)
Gender	2.07*** (0.36)	2.07*** (0.36)
AIDS/HIV	2.13*** (0.47)	2.13*** (0.47)
Bacterial infection	1.24*** (0.37)	1.23*** (0.36)
Other infection	0.87** (0.38)	0.86** (0.38)
Cancer	0.98*** (0.39)	0.97*** (0.39)
Hypertension	0.43 (0.43)	0.42 (0.43)
High cholesterol	0.22 (0.54)	0.21 (0.54)
Other heart disease	0.17 (0.40)	0.16 (0.40)
Epilepsy	1.90*** (0.49)	1.89*** (0.50)
Inflammation	−0.77 (0.63)	−0.79 (0.64)
Arthritis	0.87* (0.50)	0.87* (0.50)
Anesthetics	1.32*** (0.54)	1.31*** (0.54)
Other pain relief	0.27 (0.44)	0.28 (0.43)
Mental health	1.58*** (0.44)	1.57*** (0.45)
Asthma/allergy/respiratory	0.005 (0.41)	−0.01 (0.41)
Diabetes	0.16 (0.47)	0.15 (0.47)
Other endocrine disorders	−0.13 (0.38)	−0.14 (0.38)
Gastrointestinal disease	−0.09 (0.58)	−0.09 (0.58)
Parkinsons/MS	0.75 (0.51)	0.75 (0.51)
Glaucoma	−2.33*** (0.55)	−2.34*** (0.55)
Chronic	0.81*** (0.23)	0.81*** (0.24)
Fatal	−0.01 (0.29)	−0.01 (0.29)
Viagra	2.92*** (1.12)	2.91*** (1.12)
Dum91	0.23 (0.40)	0.22 (0.41)
Dum92	0.45 (0.46)	0.48 (0.41)
Dum93	0.21 (0.66)	0.30 (0.43)
Dum94	0.40 (0.69)	0.53 (0.50)
Dum95	0.71 (0.69)	0.85 (0.52)
Dum96	1.15* (0.63)	1.29*** (0.50)
Dum97	0.57 (0.64)	0.71 (0.52)
Dum98	1.01 (0.65)	1.16** (0.55)
Dum99	0.52 (0.65)	0.66 (0.52)
Dum00	0.49 (0.66)	0.64 (0.56)
Dum01	0.88 (0.65)	1.03* (0.56)
Constant	−1.87*** (0.53)	−1.85*** (0.55)
Dispersion	1.08	1.08

Dependent variable = ADR hospitalization count; $N = 310$. Results with PDUFA(approvals) in column (1). results with PDUFA(submissions) in column (2). Standard errors are in parentheses.*Significant at the 0.1 level.**Significant at the 0.05 level.***Significant at the 0.01 level.

Table 4
Negative binomial regression results for ADR death counts: 1990–2001 NCEs

Variable	(1)	(2)
Review time	−0.011** (0.0055)	−0.012** (0.006)
PDUFA(approvals)	−0.57 (0.62)	–
PDUFA(submissions)	–	0.04 (0.49)
Black box	0.47* (0.26)	0.46* (0.26)
Preg BBox	−0.87** (0.43)	−0.90** (0.43)
Foreign launch	0.41** (0.21)	0.43** (0.21)
Launch lag	−0.05** (0.02)	−0.046* (0.02)
Novel drug	0.99*** (0.21)	0.99*** (0.21)
Accel approval	1.03*** (0.32)	1.00*** (0.32)
log(utilization)	0.16*** (0.05)	0.16*** (0.05)
Orphan	−0.08 (0.26)	−0.05 (0.25)
Injectable	0.43* (0.23)	0.41* (0.23)
Age	0.04*** (0.004)	0.04*** (0.004)
Gender	1.27*** (0.36)	1.22*** (0.35)
AIDS/HIV	1.06** (0.55)	1.13** (0.55)
Bacterial infection	1.47*** (0.44)	1.52*** (0.44)
Other infection	0.82* (0.48)	0.88* (0.48)
Cancer	0.70 (0.47)	0.76 (0.47)
Hypertension	1.08** (0.46)	1.11** (0.46)
High cholesterol	0.31 (0.62)	0.35 (0.62)
Other heart disease	0.26 (0.46)	0.30 (0.46)
Epilepsy	1.99*** (0.56)	2.01*** (0.57)
Inflammation	−2.73*** (0.98)	−2.70*** (1.00)
Arthritis	0.79 (0.56)	0.80 (0.56)
Anesthetics	0.43 (0.64)	0.56 (0.63)
Other pain relief	−0.15 (0.56)	−0.21 (0.56)
Mental health	1.78*** (0.48)	1.80*** (0.49)
Asthma/allergy/respiratory	0.40 (0.45)	0.44 (0.46)
Diabetes	0.08 (0.53)	0.11 (0.53)
Other endocrine disorders	0.37 (0.51)	0.49 (0.50)
Gastrointestinal disease	−0.31 (0.65)	−0.29 (0.66)
Parkinsons/MS	0.30 (0.58)	0.31 (0.58)
Glaucoma	−1.79*** (0.70)	−1.75*** (0.70)
Chronic	0.84*** (0.27)	0.88*** (0.27)
Fatal	0.61** (0.32)	0.61* (0.32)
Viagra	3.98*** (1.24)	4.01*** (1.25)
Dum91	−0.84** (0.43)	−0.84** (0.43)
Dum92	0.01 (0.54)	−0.23 (0.46)
Dum93	0.03 (0.78)	−0.55 (0.47)
Dum94	0.53 (0.84)	−0.11 (0.56)
Dum95	0.74 (0.81)	0.09 (0.59)
Dum96	1.09 (0.76)	0.47 (0.57)
Dum97	0.44 (0.76)	−0.16 (0.60)
Dum98	0.96 (0.77)	0.36 (0.62)
Dum99	0.25 (0.78)	−0.37 (0.60)
Dum00	0.21 (0.78)	−0.40 (0.64)
Dum01	0.81 (0.78)	0.19 (0.64)
Constant	−1.50*** (0.58)	−1.52*** (0.60)
Dispersion	1.28	1.28

Dependent variable = ADR death count, $N = 310$. Results with PDUFA(approvals) in column (1). Results with PDUFA(submissions) in column (2). Standard errors are in parentheses.*Significant at the 0.1 level.**Significant at the 0.05 level.***Significant at the 0.01 level.

epilepsy, and mental health conditions are associated with the largest increases in ADR deaths. As in the other Tables, Viagra use is associated with a higher count of ADR deaths.

An alternative specification examined if there were any interaction effects between PDUFA and FDA review times (Hypothesis 3) or between PDUFA and drug novelty (Hypothesis 4) in the ADR regressions. The coefficients for these interaction terms reveal whether the effects of review speed or drug novelty differed in the PDUFA era. A positive coefficient is indicative of increased risk under PDUFA, while a negative coefficient is indicative of a decreased risk, perhaps due to better safety screening. Two interaction variables (PDUFA × review time; PDUFA × novel drug) were added to each ADR regression. The coefficients for the interaction term (review time × PDUFA) were not significant in any of the six ADR regressions. Hence there is little evidence for Hypothesis 3. This also suggests that increased PDUFA funds did not result in better safety screening during this period since the effect of review time was similar before and after the reform. The coefficient for the interaction term (novel drug × PDUFA) was positive and marginally significant in only one out of six ADR regressions. This provides little evidence that novel drug risks increased under PDUFA, and hence little support for Hypothesis 4.

5.3. Differences with other studies

A recent study of the speed–safety relationship using ADRs by Grabowski and Wang (2006b) did not find a significant relationship between review time and ADRs. This section discusses some key differences between our studies that may explain the different results. First, GW examine approvals from 1992 to 2002 and exclude drug approvals in 1990 and 1991 prior to PDUFA, when there was rich variation in drug review times. Differences in the set of drugs studied and periods analyzed could explain differences in the results relating to review time. GW also include both new drugs and new biologics in their study.[67] Biologics differ in important ways from drugs as described earlier, which makes them more complex and possibly risky. Variation in the review approach used in CBER and CDER along with differences in each center's workload and resources could make it difficult to pool the two in a study of the effects of review speed.

Second, in contrast to this study, GW treat accelerated approvals as mutually exclusive from a drug's priority rating. Priority or standard ratings are given to all NDAs including those submitted under the accelerated approval program, so the two accelerated approval and priority ratings are not mutually exclusive. I examined the sensitivity of my results to this assumption. If a drug's priority status is set equal to zero when it is an accelerated approval (as in GW), the coefficient for review time is negative and significant for serious ADRs and ADR hospitalizations, but became marginally significant (PDUFA approvals) and insignificant (PDUFA submission) for ADR deaths. Hence GW's treatment of accelerated approvals as mutually exclusive from a drug's priority rating may play some role in explaining differences in our results.

Third, GW use drug utilization data from IMS, which are based on prescription drug sales in retail pharmacies, hospitals, and long-term care facilities. My study uses prescription drug utilization data from the Medical Expenditure Panel Surveys (MEPSs) from 1996–2003. The MEPS data have limitations relative to IMS. One important limitation is that MEPS data are not available prior to 1996. Although sensitivity tests on a subset of the data show that the exclusion of earlier drug approvals (for which drug utilization data was not available) does not alter the results, using the 1996 data to proxy for utilization among earlier drug approvals is likely to result in higher than expected utilization for these drugs. Another limitation is that drug utilization information relating to injectable drugs, orphan drugs, and mental health drugs, may not be fully reflected in the MEPS for reasons described earlier (see p. 16, note 39, and note 62). My analysis adds three variables to control for any unmeasured utilization among drugs in these categories.[68] The MEPS prescribed medicines totals are benchmarked annually to several outside sources with similar drug data, including IMS, to make sure that MEPS estimates are within reason of similar estimates produced by other sources. However, these differences may explain why the coefficient for utilization in my study is smaller than that found in GW.

Fourth, GW exclude ADR reports from consumers. Both health professionals and consumers submit reports to the FDA. Over time the agency has made it easier for consumers to submit ADR reports and the percentage of consumer

[67] Although few biologics were in approved in the early to mid-1990s, more biologics have been approved since the late 1990s.

[68] The coefficient for log(utilization) (an elasticity) is constrained to be the same for all drugs. However, the coefficients for injectable, orphan drug, and mental health provide additional control for unmeasured utilization for drugs in each category. Coefficients for these variables are also constrained to be the same among all drugs in that category.

Table 5
The average marginal effect of a 1S.D. decline in review time on ADR counts

One standard deviation decline in review time	(1)	(2)
Dependent variable = serious ADRs		
(i) Average marginal effect (all drugs)	67	72
(ii) Average marginal effect (novel drugs)	93	101
(iii) Average marginal effect (for drugs with (above average utilization) [a]	98	106
Dependent variable = ADR hospitalizations		
(i) Average marginal effect (all drugs)	36	36
(ii) Average marginal effect (novel drugs)	48	48
(iii) Average marginal effect (for drugs with (above average utilization) [a]	60	60
Dependent variable = ADR deaths		
(i) Average marginal effect (all drugs)	11	12
(ii) Average marginal effect (novel drugs)	17	18
(iii) Average marginal effect (for drugs with (above average utilization) [a]	14	15

Column (1) reports the average marginal effects from the estimation with PDUFA(approvals) and column (2) reports marginal effects from the estimation with PDUFA(submissions). Drugs identified as outliers were not used in calculating the average marginal effects.
[a] Above average utilization is defined as 1S.D. above the mean.

reports received has increased over time. In contrast, my study includes both ADRs from health professionals and consumers. Although reports from health professionals still represent the majority of ADRs, the exclusion of ADR reports from consumers, who experience adverse effects, may bias ADR counts downward relative to my study.

Fifth, our studies consider different intervals over which ADRs are reported and different levels of aggregation for each count. GW include annual ADR counts for each drug in its first 3 years after drug launch, while my study includes a cumulative count of ADRs in the first 2 years after a drug's approval. The 2-year count in my study is a proxy for overall drug risks observed after approval. This is the period of time in which incentives to report ADRs are the greatest. In contrast, GW use three annual ADR counts per drug to proxy for drug risks in each of its first 3 years on the market. Changing rates of ADR reporting over time could complicate their approach. If incentives to report ADRs are greater in year 1 than in year 3, then the variation in drug-specific annual counts could make it difficult to detect the speed–safety relationship.

Sixth, GW include secondary suspect drugs in their counts and use a different method for aggregating adverse drug reactions. GW count an ADR report as $1/N$ in the count where N is the total number of suspect drugs (both primary and secondary) listed in the report. In contrast, my analysis counts an ADR report as 1 for the drug listed as the primary suspect drug (the drug deemed the most likely to have caused the reaction) and does not include drugs identified as a secondary suspect drugs.[69] To determine whether the estimated speed safety relationship is robust to GW's counting method, sensitivity tests were conducted using my data. When the counts include only primary suspect drugs, the results are found to be the same using the alternative counting method. Review time is negative and significant in all ADR count regressions. When counts include both primary and secondary suspect drugs as in GW ($1/N$ where N is the number of primary and secondary suspect drugs listed in the report), results showed that review time was negative and significant in the expressions for serious ADRs and ADR hospitalizations, but not for ADR deaths. This sensitivity test suggests that the use of secondary suspect drugs may increase noise in the counts making it more difficult to detect the speed–safety relationship.[70]

5.4. The marginal effect of increased review speed

Table 5 presents the average marginal effect of faster drug reviews. These marginal effects only reflect the costs of faster reviews in terms of increased ADRs, and do not include health benefits that may arise from increased review

[69] Secondary suspect drugs were first added to the ADR report in fall of 1997. Prior to 1997, reports did not include secondary suspect drugs.
[70] Such noise may arise from giving positive value to drugs not deemed the most likely to have caused the reaction; their counts value primary and secondary suspect drugs equally ($1/N$); and primary suspect drugs, which have been deemed the most likely culprit, have been valued as less than 1 in the count when there are secondary suspect drugs.

speed. The average marginal effect is given by $(1/n)\sum_{i=1}^{n}\beta_j \exp(x_i'\beta) = \beta_j\bar{y}$, where $\exp(x_i'\beta)$ is the predicted count for drug i, $\bar{y}$ is the average among the predicted values and β_j is equal to the coefficient for review time.[71] This value is then multiplied by the standard deviation for the review time variable. The average marginal effect shows the impact of a 1S.D. reduction in review time on the average predicted ADR count.[72] Average marginal effects are reported for three sets of drugs, excluding the outliers. For each ADR count row (i) in Table 5 reports the average marginal effect among all drugs. Row (ii) reports the average marginal effect of a 1S.D. reduction in review time on the ADR counts among novel drugs. Row (iii) reports the average marginal effect of a reduction in review time on the ADR counts among the set of drugs with above average utilization (defined as 2S.D. above the mean). Column (1) presents the average marginal effects for the specification using PDUFA(approvals). Column (2) presents the marginal effects from the specification using PDUFA(submissions).

Table 5 row (i) shows that a 1S.D. reduction in review time leads on average to an increase of between 67 and 72 serious ADRs in a drug's first 2 years on the market. This increase represents 21–23% of the mean serious ADR count. A 1S.D. reduction in review time increases ADR hospitalization counts on average by 36 ADRs hospitalizations in a drug's first 2 years on the market. This increase represents 19% of the mean ADR hospitalization count. A 1S.D. reduction in review time leads on average to an increase of 11–12 ADR deaths in a drug's first 2 years on the market. This increase represents 19–21% of the mean ADR death count.

The average marginal effect of increased review speed is larger among novel drugs. A 1S.D. reduction in review time leads on average to an increase of between 93 and 101 serious ADRs, an increase of 48 ADR hospitalizations, and an increase of 17–18 ADR deaths for novel drugs. These increases represent 30–32% of the mean serious ADR count, 26% of the mean ADR hospitalization count, and 30–32% of the mean ADR death count, respectively. The average marginal effect of increased review speed is also larger for drugs with above average utilization. A 1S.D. reduction in review time leads on average to an increase of 98–106 serious ADRs among high utilization drugs in the first 2 years after approval. This increase corresponds to 31–34% of the mean serious ADR count. A similar reduction in review time leads on average to an increase of 60 ADR hospitalizations and an increase of 14–15 ADR deaths for drugs with above average utilization. These increases represent 32% of the mean ADR hospitalization count and 25–26% of the mean ADR death count.

5.5. Lives lost due to increased review speed

The marginal effects are used to derive an estimate of the life years lost (from ADR deaths) due to increased review speed. To derive this estimate, the average marginal effect from Table 5 column (1) for ADR deaths is used. A 1S.D. decline in review time increases the average ADR count by 11 additional ADR deaths in its first 2 years on the market. Since 333 drugs were approved during this period, this translates into an extra 3663 ADR deaths ($11 \times 333 = 3663$) among the 1990–2001 approvals in their first 2 years on the market. The average marginal effect of increased review speed is higher among novel drugs, 17 additional ADR deaths, but lower among standard drugs, 7 additional ADR deaths. There were 139 novel drugs and 194 non-novel drugs approved in 1990–2001. Distinguishing drugs by novelty, a 1S.D. decline in review time is associated with an extra 3721 deaths $[(7 \times 194) + (17 \times 139)]$.

This value can be converted to life years lost due to ADR deaths. The mean age at ADR death is 43. The life expectancy of a 43-year-old person (using the 1997 life expectancy tables from the CDC) is 78.9. This suggests that the life years lost per ADR death are 35.9 ($78.9 - 43 = 35.9$). Multiplying the life years lost per ADR death by the number of ADR deaths yields an estimate of 131501.7 life years lost from extra ADR deaths (in the first 2 years after approval) due to increased FDA review speed. With the distinction for drug novelty, this value increases to 133583.9 life years lost from the extra ADR deaths associated with increased FDA review speed. If individuals with

[71] The average marginal effect will be larger than the marginal effect for a representative drug (a drug with characteristics equal to their sample means) because of the convexity of the exponential mean function. Cameron and Trivedi (1998) suggest that it is conceptually better to report the former.

[72] Another possible benchmark for the marginal effects is 14.3 months, the difference between the average review time in 1990–1992 and 1999–2001 ($31 - 16.7$ months). With this benchmark, marginal effects would be equal to 0.75 times the values listed in Table 5(since $14.3 = .75$ (standard deviation of review time)).

some chronic illnesses have shorter than expected life spans, then this would result in an upward bias in this cost estimate.[73]

The cost estimate is subject to three downward biases. First, only ADR deaths in the first 2 years on the market are considered. Additional deaths beyond that period are not included in the above estimate. Second, ADRs are underreported, and hence the above estimate does not reflect all ADR deaths or the life years lost from increased review speed.[74] Third, the above estimate does not consider the impact of other serious ADRs and ADR hospitalizations. Although other serious ADRs do not result in death, they could reduce quality of life or patient life years as a consequence of the severity and duration of the adverse effect. Since these other serious ADRs are more frequent than ADR deaths, finding a way to quantify their impact on patient health remains a topic for future research.

Also important are the benefits to patients from gaining faster access to useful new drugs. Relying on drug sales as a measure of benefits, Philipson et al. (2005) suggest that PDUFA raised producer and consumer surplus by $18–31 billion, which they argue translates into 180,000–210,000 life years gained for patients. Although sales are a good indicator of industry benefit, sales provide little information about the gains in patient health derived from faster access to a new drug instead of an older one. For instance, a drug that is ineffective, equivalent to existing drugs, or associated with serious side effects leading to hospitalization, disability, or death would be counted as a benefit using their approach. Sales will also reflect high drug prices, which also need not correlate with health benefits. These factors are likely to result in an upward bias in their estimated health benefits of PDUFA. Additional research is needed to quantify the health benefits of faster drug access.

6. Conclusions

This paper assesses the effects of FDA drug review times and user fees on drug-related risks among the 1990–2001 drug approvals. The primary result is that drugs receiving faster FDA reviews have higher counts of serious adverse drug reactions, including those resulting in hospitalization and death. A 1S.D. reduction in review time is associated with a 21–23% increase in serious ADRs, a 19% increase in ADR hospitalizations and a 19–21% increase in ADR deaths. The magnitude of this effect is greater for novel drugs and drugs with above average utilization. Furthermore, this effect is not explained by differences in the conditions treated or differences in characteristics of the drugs studied. Although any increase in risks from ADRs must be weighed against a drug's benefits, the results suggest that there is a trade-off between review speed and drug safety. Additional evidence shows that after controlling for the effects of review speed, PDUFA approvals and PDUFA submissions show little association with ADRs. These results suggest PDUFA's impact on drug-related risks occurs primarily through the speed of new drug review.

These findings differ from previous studies of drug withdrawals or black box warnings, which found no change in drug safety over time (Berndt et al., 2005; Begosh et al., 2006). By using a richer measure of drug safety that does not reflect regulatory discretion, this study finds that increased review speed has led to more drug-related risks. This finding is consistent with prior results by Olson (2002) and Olson (2004b), who studied the 1990–1995 drug approvals. The results differ from Grabowski and Wang (2006b), who did not find an association between review times and ADRs for the 1992–2002 approvals. In contrast to my study, GW exclude 1990 and 1991 approvals and they include both drugs and biologics together. Differences in the set of drugs studied and periods analyzed could explain their different findings. However, differences in the kinds of ADRs considered and the way in which ADRs are counted in their study could also play a role.[75] GW exclude ADR reports from consumers, which may contain valuable risk information about the adverse effects experienced by consumers. Their analysis also includes secondary suspect drugs in addition to the primary suspect drug, which may increase noise in their counts.[76] More noisy ADR counts could also explain why GW could not find an association between speed and safety.

[73] Using the marginal effect for a representative drug, the less preferred measure, may provide a lower bound on the estimate. A 1S.D. reduction in review time is associated with four extra ADR deaths for a representative drug. This yields an estimate of 45,054 life years lost from extra ADR deaths due to increased review speed over time.

[74] For example, if there is 30% underreporting, then the estimate increases to 187859.6 life years lost from the ADR deaths associated with faster drug reviews. Including the distinction between novel and standard drugs, it increases to 190834.1 life years lost from the ADR deaths arising from faster FDA drug reviews.

[75] Both studies include controls for launch lag, black box warnings, and drug novelty. GW's study does not include a dummy variable for drug's first launched abroad.

[76] GW give equal weight to drugs deemed to be the primary suspect and secondary drugs in the counts.

Additional results show that drugs given a priority rating by the FDA, accelerated approvals, drugs receiving a black box warning at the time of approval, drugs launched abroad prior to US approval, and drugs having shorter US launch lags are also associated with more ADRs. Even after controlling for these other important influences on ADR counts, there is still evidence of a negative and significant relationship between FDA review speed and drug safety. This evidence supports the hypothesis that increases in the speed of drug review have increased drug-related risks over time.

Key study limitations include the following. First, there is no estimate of the health benefits of increased review speed. Previous research (Lichtenberg, 2005; Olson, 2004b; Philipson et al., 2005) suggests that faster access to novel drugs, in particular, is likely to generate high health benefits. Such health benefits can be expected to vary among patients and conditions. For instance patients with life-threatening diseases such as AIDs, Alzheimers, and Parkinsons, have much at stake in obtaining faster access to new innovative drugs and may be willing to bear more risk than others who have less serious ailments for which there are other safe and effective remedies. This suggests that the risk benefit assessment of increased review speed and PDUFA will vary by drug class, drug novelty, and condition severity. Second, this study does not include drug approvals beyond 2002. It is not known whether the speed–safety relationship extends to these drug approvals.[77] Third, this study uses 1996 data to proxy for drug utilization for drugs approved in 1990–1993 since the MEPS drug utilization data were not available before 1996. Hence, utilization may be overstated for these drug approvals relative to other drugs. Tests on a subset of the data, which excluded these drugs, showed that results were not sensitive to this assumption.

Given that speedy FDA drug reviews have led to increased drug-related risks, how should policymakers respond? Should regulators slow down the review process, particularly for drugs deemed to be more risky? Results from this study cannot fully answer this question. The reason is that slowing down the review process can have other adverse consequences on patients that may offset the reduction in risks from ADRs. This includes health benefits that are lost due to reduced drug access for patients.[78] The answer depends on how the risk of ADRs compares to the health benefits from greater drug access. For example, novel drugs are more risky, but they also offer health benefits over existing drugs. Slower reviews would limit patient access to those health benefits. Similarly, regulators recognize that the drugs approved with boxed warnings have serious risks, but the decision to approve suggests that regulators believe the benefits outweigh those risks. Information about both the costs and the benefits of faster drug access are needed to fully evaluate the effects of increased review speed and PDUFA.

In addition to review speed, however, several factors including priority ratings, accelerated approvals, foreign drug launch, and shorter US launch lags have also led to increased ADRs for patients. Are there other policy responses to this problem of increased ADRs? Improving the system of post-marketing drug safety surveillance can lead to faster detection of new risks that emerge after approval and more timely responses to drug safety problems. Better communication of drug risk information to patients and physicians in the post-market can also lead to more informed drug choices and better patient/drug matches. Results from this analysis suggest that the value of post-marketing resources devoted to these activities has increased in the PDUFA era. This is consistent with current calls from the IOM (2006) and Congress to increase the resources (from PDUFA funding) and authority for post-marketing drug safety activities and risk communication. Although such policies will not eliminate ADR risks, they will help reduce adverse effects over time without sacrificing any of the health gains from faster drug access.

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[77] Since regulators have become increasingly effective in meeting PDUFA drug review deadlines, analyses that use more recent drug approvals could be complicated by smaller variation in review times.

[78] Vernon et al. (2006) also identifies reduced incentives for R&D affecting the future availability of new medicines as another indirect cost of longer reviews.

Appendix A. First stage regression results (for exogeneity test)

Variable	Coefficient	Interaction terms (with PDUFA)
Firm characteristics (instruments)		
R&D/sales	−119.67*** (34.58)	119.32*** (34.60)
Drug sales/total sales	−3.75 (5.75)	4.55 (6.81)
Current experience	−3.46*** (1.08)	2.72* (1.63)
Past approvals (since 1990)	2.96* (1.59)	−2.98* (1.67)
log(employees)	−2.76 (1.91)	0.76 (2.14)
US firm	7.07* (4.29)	−4.33 (4.75)
New firm	−46.38*** (14.22)	39.08*** (14.99)
Other exogenous variables in (1)	*Included*	
Condition effects	Included	
Fixed time effects	Included	
N	298	
Adjusted R^2	0.75	
F (on excluded instruments) [a]	2.37	

Dependent variable = review time. White corrected standard errors are in parentheses.*Significant at the 0.1 level.**Significant at the 0.05 level.***Significant at the 0.01 level.

[a] The F-statistic suggests weak correlation with the excluded instruments. Such correlation could pose challenges for the purpose of IV estimation if it were used.

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